

Lab Experiments
Set:16-20
Date:29-07-2026

Set 16: Customer Demographics Analysis

1. Bar Chart (Customer Age Distribution)

Code:

```
customer <- data.frame(  
  CustomerID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)  
barplot(customer$Age,  
  names.arg = customer$CustomerID,  
  col = "skyblue",  
  main = "Customer Age Distribution",  
  xlab = "Customer ID",  
  ylab = "Age (Years)")
```

Output:



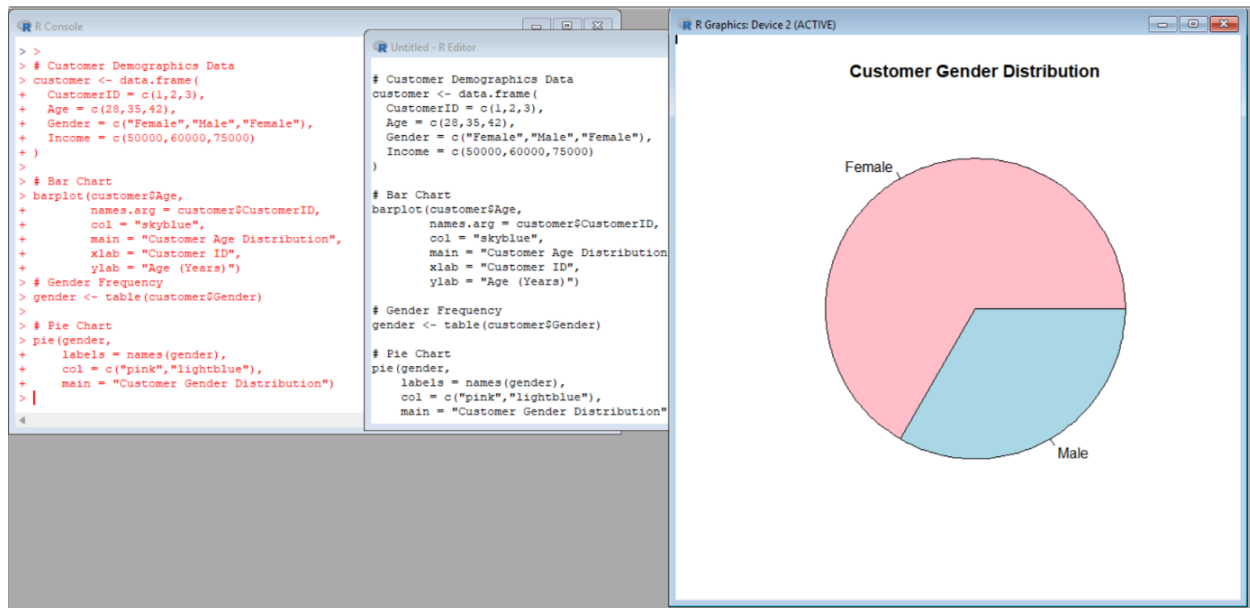
2: Pie Chart (Gender Distribution)

Code:

```
gender <- table(customer$Gender)  
pie(gender,  
  labels = names(gender),
```

```
col = c("pink","lightblue"),
main = "Customer Gender Distribution")
```

Output:



3: Table

Code:

```
customer <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
print(customer)
```

Output:

```
>
> print(customer)
  CustomerID Age Gender Income
1          1  28 Female  50000
2          2  35  Male  60000
3          3  42 Female  75000
> |
```

4: Dashboard (R Shiny)

Code:

```
library(shiny)
customer <- data.frame(
```

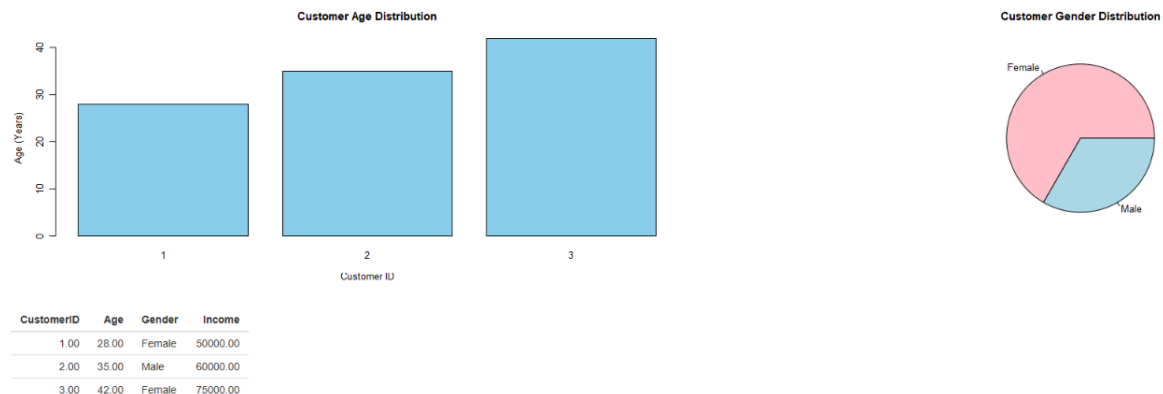
```

CustomerID = c(1,2,3),
Age = c(28,35,42),
Gender = c("Female","Male","Female"),
Income = c(50000,60000,75000)
)
ui <- fluidPage(
  titlePanel("Customer Demographics Dashboard"),
  fluidRow(
    column(6, plotOutput("bar")),
    column(6, plotOutput("pie"))
  ),
  br(),
  tableOutput("table")
)
server <- function(input, output){
  output$bar <- renderPlot({
    barplot(customer$Age,
      names.arg = customer$CustomerID,
      col = "skyblue",
      main = "Customer Age Distribution",
      xlab = "Customer ID",
      ylab = "Age (Years)")
  })
  output$pie <- renderPlot({
    gender <- table(customer$Gender)
    pie(gender,
      labels = names(gender),
      col = c("pink","lightblue"),
      main = "Customer Gender Distribution")
  })
  output$table <- renderTable({
    customer
  })
}
shinyApp(ui, server)

```

Output:

Customer Demographics Dashboard



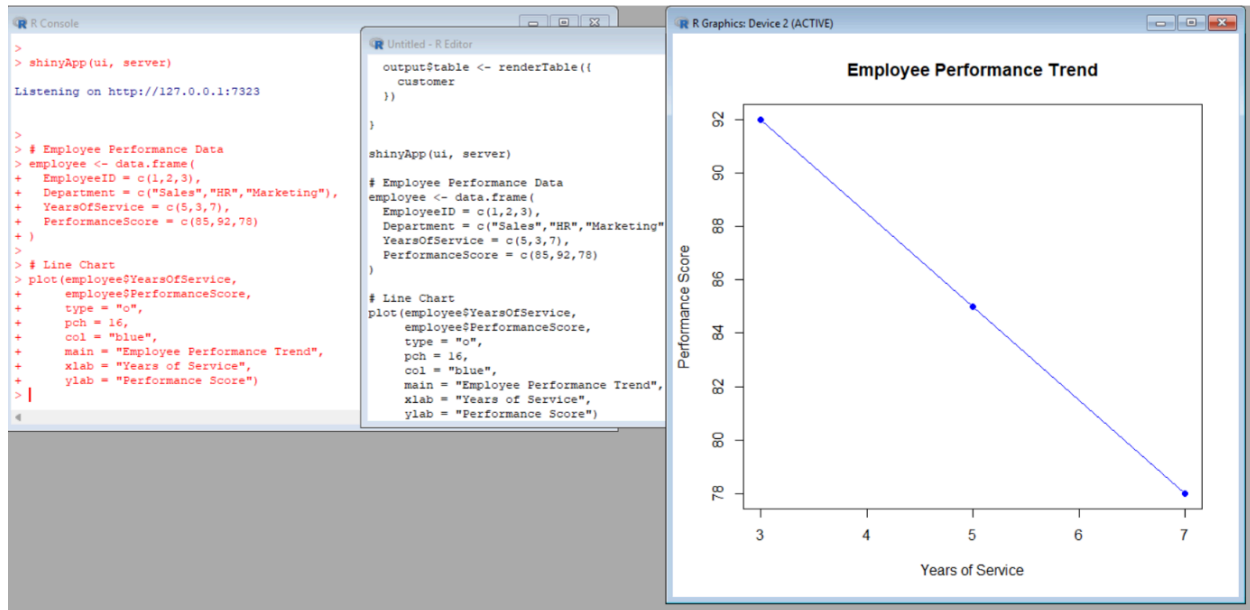
Set 17: Employee Performance Analysis

1: Line Chart (Performance Trend)

Code:

```
employee <- data.frame(  
  EmployeeID = c(1,2,3),  
  Department = c("Sales","HR","Marketing"),  
  YearsOfService = c(5,3,7),  
  PerformanceScore = c(85,92,78)  
)  
plot(employee$YearsOfService,  
  employee$PerformanceScore,  
  type = "o",  
  pch = 16,  
  col = "blue",  
  main = "Employee Performance Trend",  
  xlab = "Years of Service",  
  ylab = "Performance Score")
```

Output:

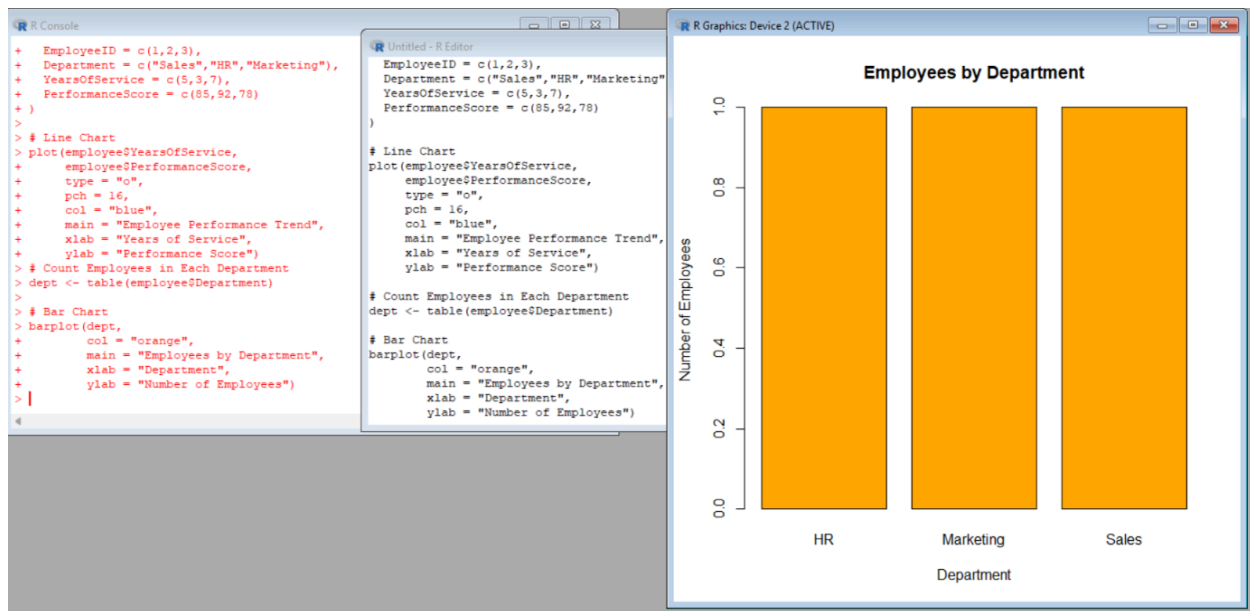


2: Bar Chart (Department Distribution)

Code:

```
dept <- table(employee$Department)
barplot(dept,
        col = "orange",
        main = "Employees by Department",
        xlab = "Department",
        ylab = "Number of Employees")
```

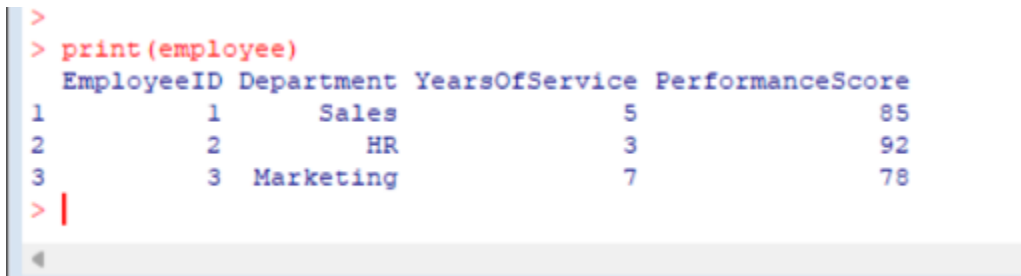
Output:



3: Table

Code:

```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
print(employee)
```

Output:


```
>
> print(employee)
  EmployeeID Department YearsOfService PerformanceScore
1           1      Sales              5                85
2           2         HR              3                92
3           3  Marketing              7                78
> |
```

4: Dashboard (R Shiny)**Code:**

```
library(shiny)
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  fluidRow(
    column(6, plotOutput("line")),
    column(6, plotOutput("bar"))
  ),
  br(),
  tableOutput("table")
)
server <- function(input, output){
  output$line <- renderPlot({
    plot(employee$YearsOfService,
          employee$PerformanceScore,
          type = "o",
          pch = 16,
```

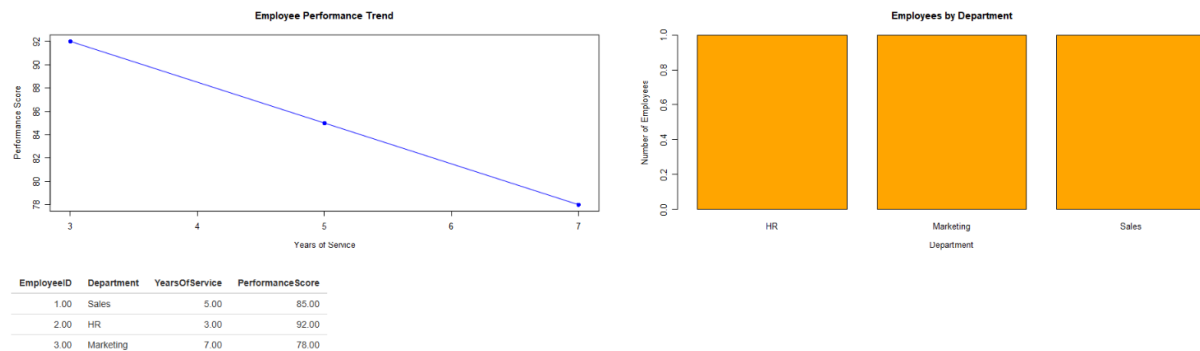
```

    col = "blue",
    main = "Employee Performance Trend",
    xlab = "Years of Service",
    ylab = "Performance Score")
  })
  output$bar <- renderPlot({
    dept <- table(employee$Department)
    barplot(dept,
      col = "orange",
      main = "Employees by Department",
      xlab = "Department",
      ylab = "Number of Employees")
  })
  output$table <- renderTable({
    employee
  })
}
shinyApp(ui, server)

```

Output:

Employee Performance Dashboard



Set 18: Product Inventory Management

1: Bar Chart (Quantity Available)

Code:

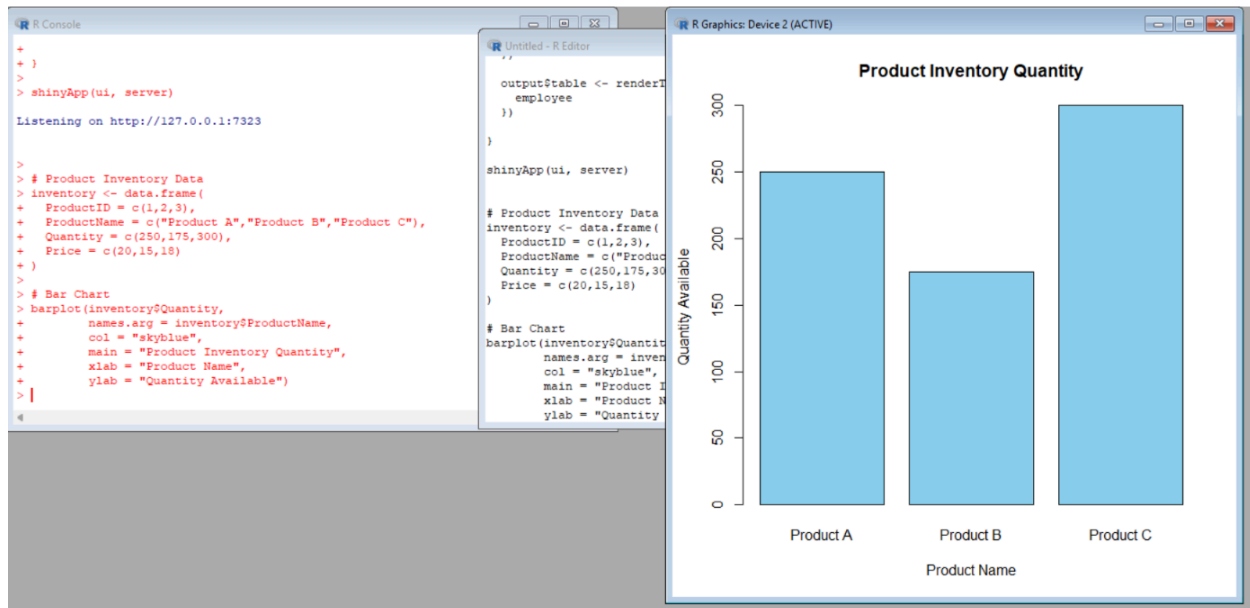
```

inventory <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  Quantity = c(250,175,300),
  Price = c(20,15,18)
)

```

```
)
barplot(inventory$Quantity,
        names.arg = inventory$ProductName,
        col = "skyblue",
        main = "Product Inventory Quantity",
        xlab = "Product Name",
        ylab = "Quantity Available")
```

Output:

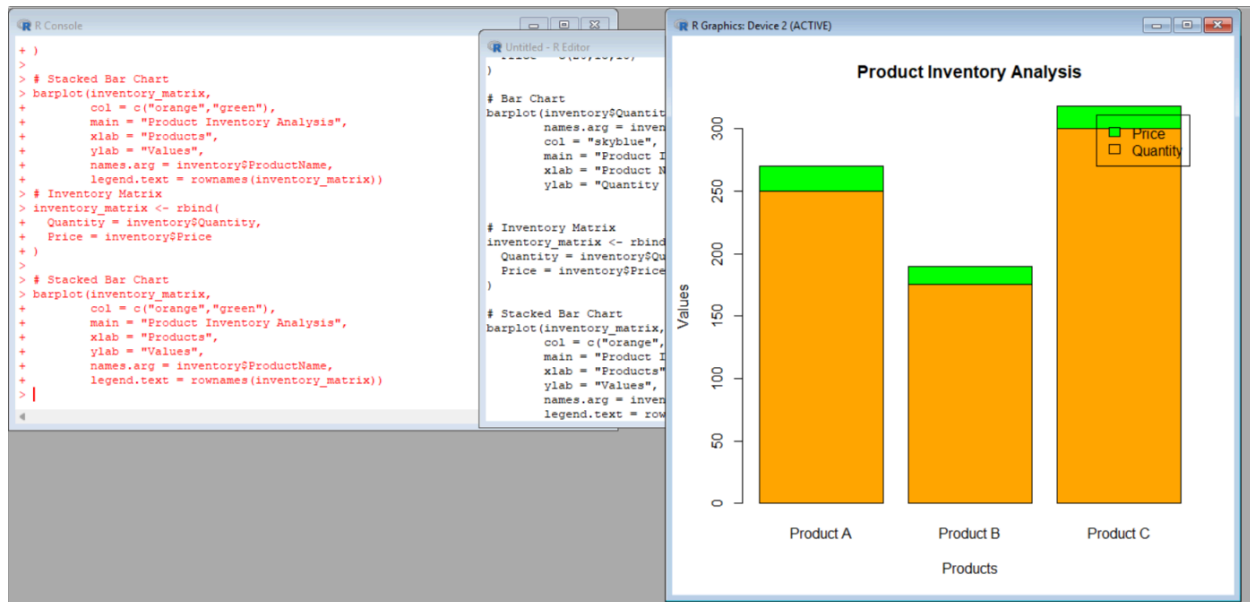


2: Stacked Bar Chart

Code:

```
inventory_matrix <- rbind(
  Quantity = inventory$Quantity,
  Price = inventory$Price
)
barplot(inventory_matrix,
        col = c("orange", "green"),
        main = "Product Inventory Analysis",
        xlab = "Products",
        ylab = "Values",
        names.arg = inventory$ProductName,
        legend.text = rownames(inventory_matrix))
```

Output:



3: Table

Code:

```

inventory <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  Quantity = c(250,175,300),
  Price = c(20,15,18)
)
print(inventory)

```

Output:

```

>
> print(inventory)
  ProductID ProductName Quantity Price
1         1   Product A      250    20
2         2   Product B      175    15
3         3   Product C      300    18
>

```

4: Dashboard (R Shiny)

Code:

```

library(shiny)
inventory <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),

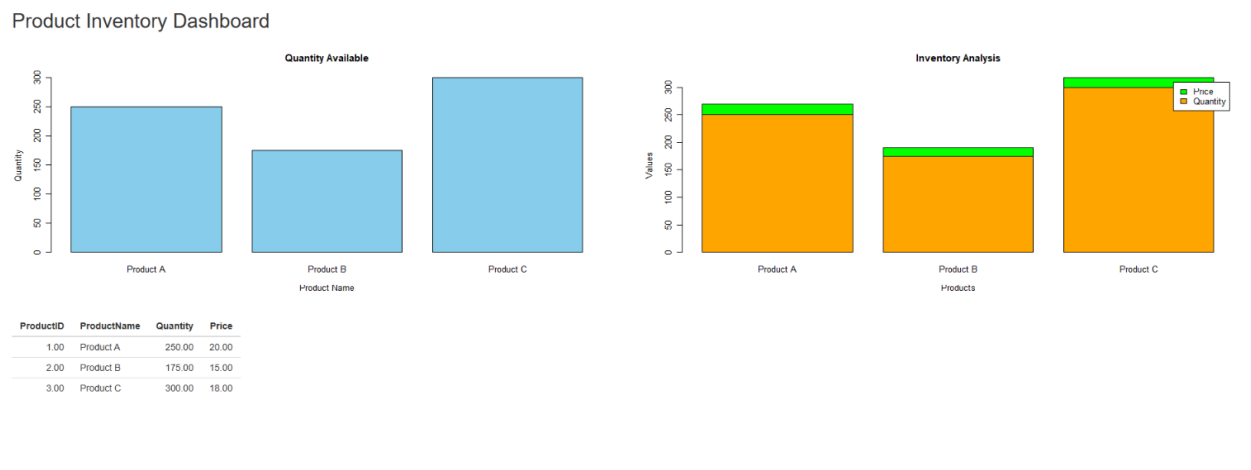
```

```

    Quantity = c(250,175,300),
    Price = c(20,15,18)
  )
  ui <- fluidPage(
    titlePanel("Product Inventory Dashboard"),
    fluidRow(
      column(6, plotOutput("bar")),
      column(6, plotOutput("stacked"))
    ),
    br(),
    tableOutput("table")
  )
  server <- function(input, output){
    output$bar <- renderPlot({
      barplot(inventory$Quantity,
        names.arg = inventory$ProductName,
        col = "skyblue",
        main = "Quantity Available",
        xlab = "Product Name",
        ylab = "Quantity")
    })
    output$stacked <- renderPlot({
      inventory_matrix <- rbind(
        Quantity = inventory$Quantity,
        Price = inventory$Price
      )
      barplot(inventory_matrix,
        col = c("orange","green"),
        main = "Inventory Analysis",
        xlab = "Products",
        ylab = "Values",
        names.arg = inventory$ProductName,
        legend.text = rownames(inventory_matrix))
    })
    output$table <- renderTable({
      inventory
    })
  }
  shinyApp(ui, server)

```

Output:



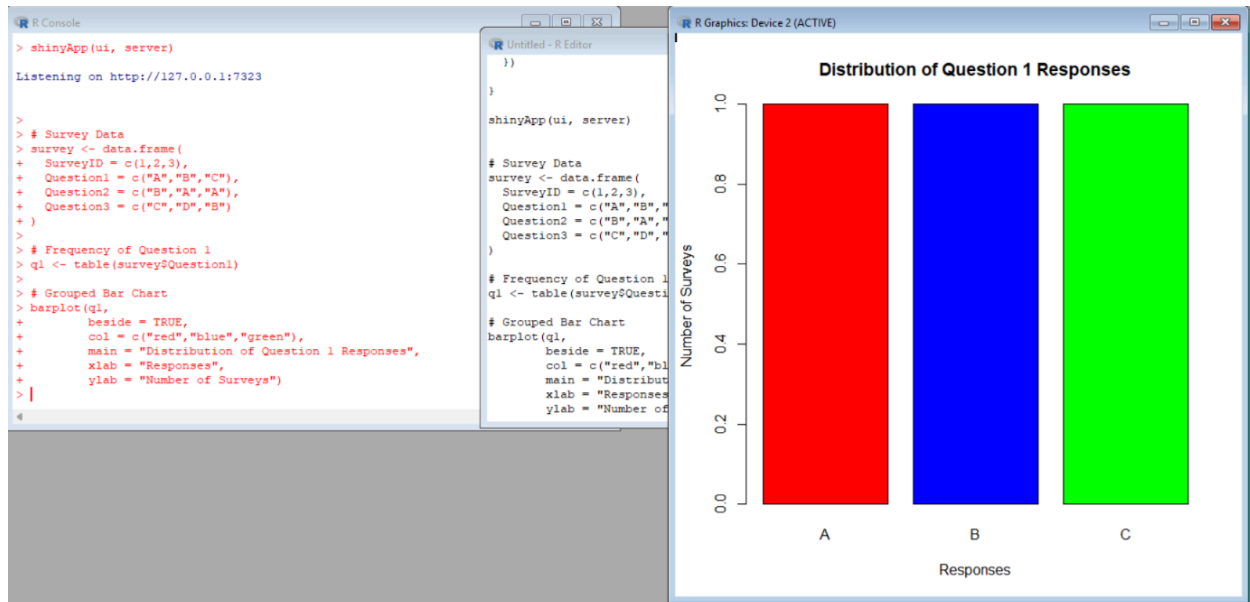
Set 19: Survey Responses Analysis

1: Grouped Bar Chart (Question 1 Responses)

Code:

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
q1 <- table(survey$Question1)
barplot(q1,
  beside = TRUE,
  col = c("red","blue","green"),
  main = "Distribution of Question 1 Responses",
  xlab = "Responses",
  ylab = "Number of Surveys")
```

Output:



2: Stacked Bar Chart (All Questions)

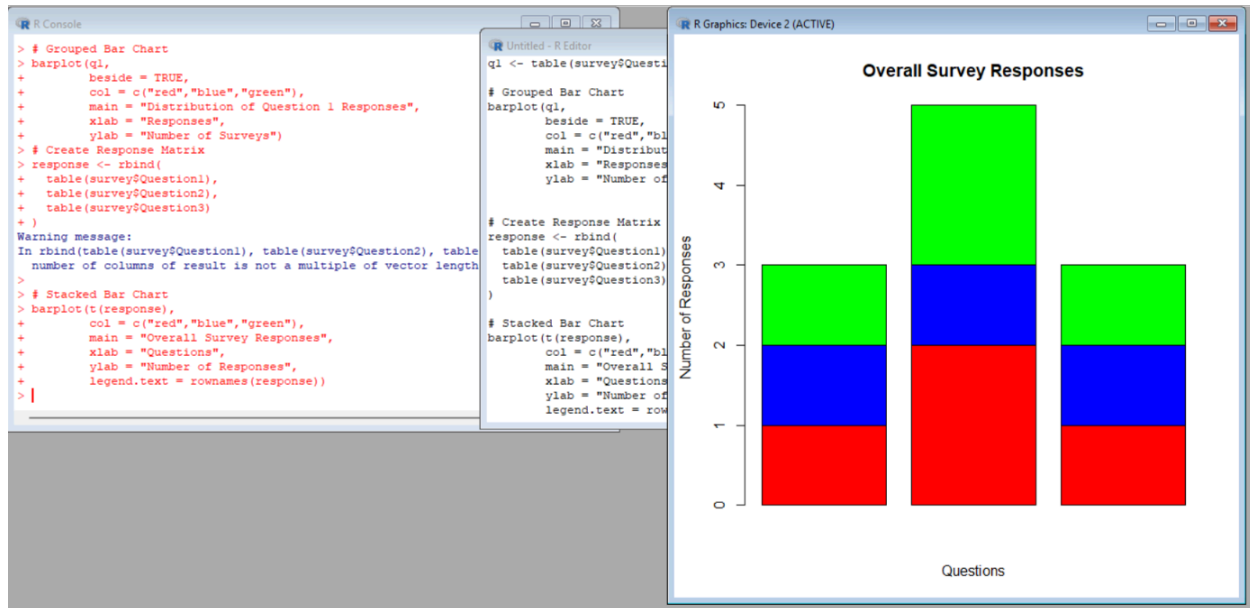
Code:

```

response <- rbind(
  table(survey$Question1),
  table(survey$Question2),
  table(survey$Question3)
)
barplot(t(response),
  col = c("red","blue","green"),
  main = "Overall Survey Responses",
  xlab = "Questions",
  ylab = "Number of Responses",
  legend.text = rownames(response))

```

Output:



3: Table

Code:

```
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
print(survey)
```

Output:

```
+ )
>
> print(survey)
  SurveyID Question1 Question2 Question3
1         1         A         B         C
2         2         B         A         D
3         3         C         A         B
> |
```

4: Dashboard (R Shiny)

Code:

```
library(shiny)
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
```

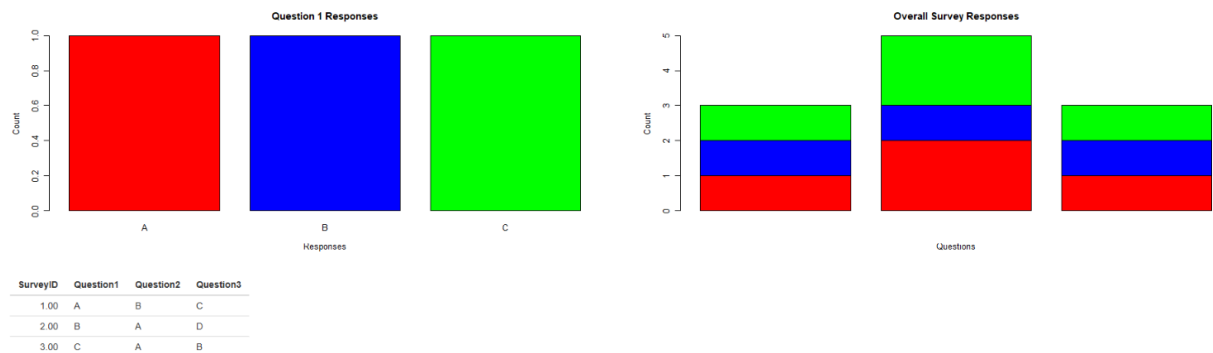
```

Question2 = c("B", "A", "A"),
Question3 = c("C", "D", "B")
)
ui <- fluidPage(
  titlePanel("Survey Responses Dashboard"),
  fluidRow(
    column(6, plotOutput("groupbar")),
    column(6, plotOutput("stackbar"))
  ),
  br(),
  tableOutput("table")
)
server <- function(input, output){
  output$groupbar <- renderPlot({
    q1 <- table(survey$Question1)
    barplot(q1,
      beside = TRUE,
      col = c("red", "blue", "green"),
      main = "Question 1 Responses",
      xlab = "Responses",
      ylab = "Count")
  })
  output$stackbar <- renderPlot({
    response <- rbind(
      table(survey$Question1),
      table(survey$Question2),
      table(survey$Question3)
    )
    barplot(t(response),
      col = c("red", "blue", "green"),
      main = "Overall Survey Responses",
      xlab = "Questions",
      ylab = "Count",
      legend.text = rownames(response))
  })
  output$table <- renderTable({
    survey
  })
}
shinyApp(ui, server)

```

Output:

Survey Responses Dashboard



Set 20: Stock Analysis

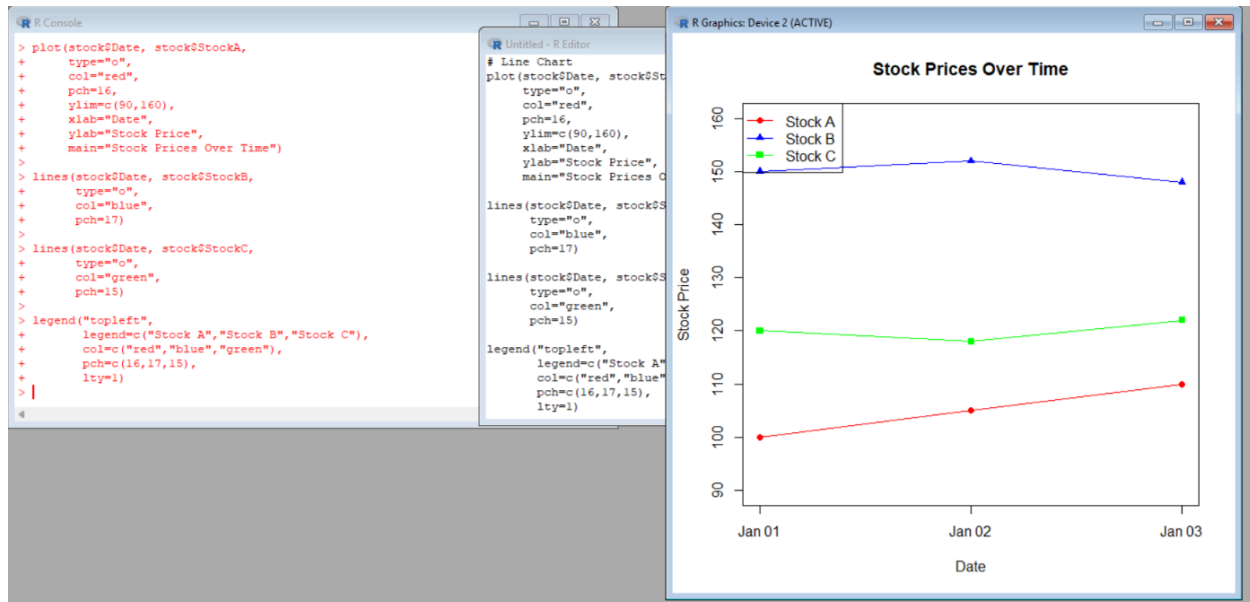
1: Line Chart (Stock Prices)

Code:

```
stock <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
  StockA = c(100,105,110),
  StockB = c(150,152,148),
  StockC = c(120,118,122)
)
plot(stock$Date, stock$StockA,
  type="o",
  col="red",
  pch=16,
  ylim=c(90,160),
  xlab="Date",
  ylab="Stock Price",
  main="Stock Prices Over Time")
lines(stock$Date, stock$StockB,
  type="o",
  col="blue",
  pch=17)
lines(stock$Date, stock$StockC,
  type="o",
  col="green",
  pch=15)
legend("topleft",
  legend=c("Stock A","Stock B","Stock C"),
  col=c("red","blue","green"),
```

```
pch=c(16,17,15),
lty=1)
```

Output:

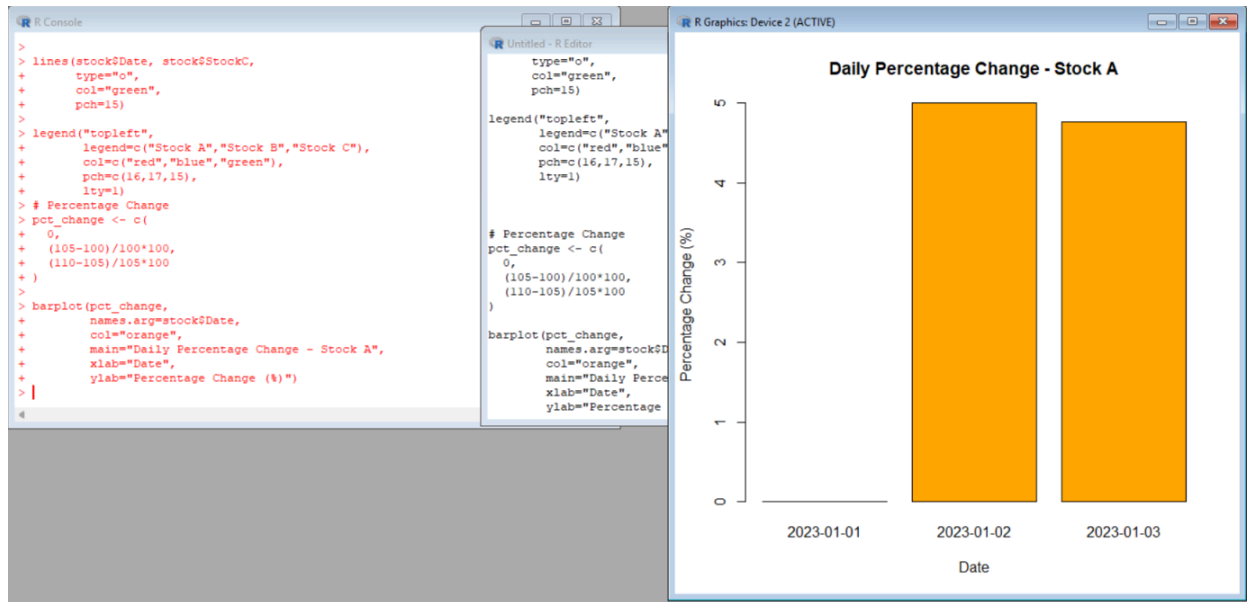


2: Bar Chart (Daily Percentage Change – Stock A)

Code:

```
pct_change <- c(
  0,
  (105-100)/100*100,
  (110-105)/105*100
)
barplot(pct_change,
        names.arg=stock$Date,
        col="orange",
        main="Daily Percentage Change - Stock A",
        xlab="Date",
        ylab="Percentage Change (%)")
```

Output:



3. Table

Code:

```
stock <- data.frame(
  Date = c("2023-01-01", "2023-01-02", "2023-01-03"),
  StockA = c(100, 105, 110),
  StockB = c(150, 152, 148),
  StockC = c(120, 118, 122)
)
print(stock)
```

Output:

```
+ )
>
> print(stock)
  Date StockA StockB StockC
1 2023-01-01   100   150   120
2 2023-01-02   105   152   118
3 2023-01-03   110   148   122
>
```

4: Dashboard (R Shiny)

Code:

```
library(shiny)
stock <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  StockA = c(100, 105, 110),
  StockB = c(150, 152, 148),
```

```

    StockC = c(120,118,122)
  )
  ui <- fluidPage(
    titlePanel("Stock Analysis Dashboard"),
    fluidRow(
      column(6, plotOutput("line")),
      column(6, plotOutput("bar"))
    ),
    br(),
    tableOutput("table")
  )
  server <- function(input, output){
    output$line <- renderPlot({
      plot(stock$Date, stock$StockA,
        type="o",
        col="red",
        pch=16,
        ylim=c(90,160),
        xlab="Date",
        ylab="Stock Price",
        main="Stock Prices")
      lines(stock$Date, stock$StockB,
        type="o",
        col="blue",
        pch=17)
      lines(stock$Date, stock$StockC,
        type="o",
        col="green",
        pch=15)
      legend("topleft",
        legend=c("Stock A","Stock B","Stock C"),
        col=c("red","blue","green"),
        pch=c(16,17,15),
        lty=1)
    })
    output$bar <- renderPlot({
      pct_change <- c(
        0,
        (105-100)/100*100,
        (110-105)/105*100

```

```

)
barplot(pct_change,
       names.arg=stock$Date,
       col="orange",
       main="Stock A Percentage Change",
       xlab="Date",
       ylab="Percentage Change (%)")
})
output$table <- renderTable({
  stock
})
}
shinyApp(ui, server)

```

Output:

